

- Description

This module manages a smart gas leak detection system, ensuring that:

1. The system continuously monitors the concentration of flammable gas in the air (expressed as a percentage of the lower explosive limit, LEL).
2. If the gas concentration exceeds a predefined warning threshold, a warning alert is triggered.
3. If the gas concentration exceeds a critical danger threshold, an emergency shutdown is initiated, and a high-risk alarm is triggered.
4. If the gas concentration remains within safe limits, no action is taken.

This system helps enhance safety by providing early detection of potential gas leaks and preventing hazardous situations.

- Specification

module SmartGasLeakDetection

var currentGasLevel, warningThreshold, dangerThreshold: real

var emergencyShutdown: boolean

process updateGasLevel(newGasLevel: real) returns (currentGasLevel: real, emergencyShutdown: boolean)

functional scenario form:

currentGasLevel > dangerThreshold && emergencyShutdown = true, output "ALERT: Critical gas leak detected (" + currentGasLevel + "% LEL). Emergency shutdown initiated.", Emergency shutdown activated.

||

currentGasLevel <= dangerThreshold && emergencyShutdown = false, output "WARNING: Gas level high (" + currentGasLevel + "% LEL). Immediate attention required.", Gas warning alert triggered.

||

currentGasLevel <= warningThreshold && emergencyShutdown = false, output "Gas concentration is within safe limits (" + currentGasLevel + "% LEL).", Gas level is within safe range.

end process

end module

- Java Code

```
package code;
public class c113_SmartGasLeakDetector {
    private double warningThreshold;
    private double dangerThreshold;
    private double currentGasLevel;
    private boolean emergencyShutdown;
    public c113_SmartGasLeakDetector(double warningThreshold, double dangerThreshold, double initialGasLevel) {
        this.warningThreshold = warningThreshold;
```

```

        this.dangerThreshold = dangerThreshold;
        this.currentGasLevel = initialGasLevel;
        this.emergencyShutdown = false;
    }
    public void updateGasLevel(double newGasLevel) {
        currentGasLevel = newGasLevel;
        checkGasLeak();
        System.out.printf("Current Gas Level: %.2f%% LEL, Emergency Shutdown: %s\n",
                           currentGasLevel, emergencyShutdown ? "ACTIVE" : "INACTIVE");
        if (emergencyShutdown) {
            System.out.println("ALERT: Critical gas leak detected! Emergency shutdown initiated.");
        } else if (currentGasLevel > warningThreshold) {
            System.out.println("WARNING: Gas level high. Immediate attention required.");
        } else {
            System.out.println("Gas concentration is within safe limits.");
        }
    }
    private void checkGasLeak() {
        if (currentGasLevel > dangerThreshold) {
            emergencyShutdown = true;
        } else {
            emergencyShutdown = false;
        }
    }
    public double getCurrentGasLevel() {
        return currentGasLevel;
    }
    public static void main(String[] args) {
        c113_SmartGasLeakDetector detector = new c113_SmartGasLeakDetector(20.0, 40.0, 10.0);
        // testcase-VT:
        //     detector.updateGasLevel(15.0);
        //     detector.updateGasLevel(25.0);
        //     detector.updateGasLevel(45.0);

        // testcase-FT:
        //     detector.updateGasLevel(6.365249299686023E293);
        //     detector.updateGasLevel(1.6349709418800172E296);
        //     detector.updateGasLevel(2.3630739519577067E295);
    }
}

```

```

        // testcase-Z3:
        //     detector.updateGasLevel(15.0);
        //     detector.updateGasLevel(25.0);
        //     detector.updateGasLevel(45.0);

        // testcase-UVT:
        detector.updateGasLevel(15.0);
        detector.updateGasLevel(25.0);
        detector.updateGasLevel(45.0);
    }
}

```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c113_SmartGasLeakDetector.java	100.0 %	8	0	8
▼ c113_SmartGasLeakDetector	100.0 %	8	0	8
● updateGasLevel(double)	100.0 %	6	0	6
■ checkGasLeak()	100.0 %	2	0	2
● c113_SmartGasLeakDetector(do		0	0	0
● main(String[])		0	0	0

3. Vibration testing instruction coverage: 96.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,997	22,090
▼ src	0.4 %	93	21,997	22,090
▼ code	0.4 %	93	21,997	22,090
▼ c113_SmartGasLeakDetector.java	96.9 %	93	3	96
▼ c113_SmartGasLeakDetector	96.9 %	93	3	96
● updateGasLevel(double)	100.0 %	47	0	47
■ checkGasLeak()	100.0 %	14	0	14
● c113_SmartGasLeakDetector(do	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

4. Fuzz testing branch coverage: 37.5%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.4 %	3	839	842
▼ src	0.4 %	3	839	842
▼ code	0.4 %	3	839	842
▼ c113_SmartGasLeakDetector.java	37.5 %	3	5	8
▼ c113_SmartGasLeakDetector	37.5 %	3	5	8
● updateGasLevel(double)	33.3 %	2	4	6
■ checkGasLeak()	50.0 %	1	1	2
● c113_SmartGasLeakDetector(do		0	0	0
● main(String[])		0	0	0

5. Fuzz testing instruction coverage: 79.2%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	76	22,014	22,090
▼ src	0.3 %	76	22,014	22,090
▼ code	0.3 %	76	22,014	22,090
▼ c113_SmartGasLeakDetector.java	79.2 %	76	20	96
▼ c113_SmartGasLeakDetector	79.2 %	76	20	96
● updateGasLevel(double)	70.2 %	33	14	47
■ checkGasLeak()	78.6 %	11	3	14
● c113_SmartGasLeakDetector(do	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c113_SmartGasLeakDetector.java	100.0 %	8	0	8
▼ c113_SmartGasLeakDetector	100.0 %	8	0	8
● updateGasLevel(double)	100.0 %	6	0	6
■ checkGasLeak()	100.0 %	2	0	2
● c113_SmartGasLeakDetector(do		0	0	0
● main(String[])		0	0	0

7. Z3 instruction coverage: 96.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,997	22,090
▼ src	0.4 %	93	21,997	22,090
▼ code	0.4 %	93	21,997	22,090
▼ c113_SmartGasLeakDetector.java	96.9 %	93	3	96
▼ c113_SmartGasLeakDetector	96.9 %	93	3	96
● updateGasLevel(double)	100.0 %	47	0	47
■ checkGasLeak()	100.0 %	14	0	14
● c113_SmartGasLeakDetector(do	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	1.0 %	8	834	842
▼ src	1.0 %	8	834	842
▼ code	1.0 %	8	834	842
▼ c113_SmartGasLeakDetector.java	100.0 %	8	0	8
▼ c113_SmartGasLeakDetector	100.0 %	8	0	8
● updateGasLevel(double)	100.0 %	6	0	6
■ checkGasLeak()	100.0 %	2	0	2
● c113_SmartGasLeakDetector(do		0	0	0
● main(String[])		0	0	0

9. Unimproved V-method instruction coverage: 96.9%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	93	21,997	22,090
▼ src	0.4 %	93	21,997	22,090
▼ code	0.4 %	93	21,997	22,090
▼ c113_SmartGasLeakDetector.java	96.9 %	93	3	96
▼ c113_SmartGasLeakDetector	96.9 %	93	3	96
● updateGasLevel(double)	100.0 %	47	0	47
■ checkGasLeak()	100.0 %	14	0	14
● c113_SmartGasLeakDetector(do	100.0 %	15	0	15
● main(String[])	100.0 %	17	0	17